

Purr-fectly Average: Possible or Impossible?

The Cat Café Census · based on the NRICH problem *About Average*



Name: _____

Date: _____

Partner: _____

At the Cat Café, every cat counts the mice it catches. Today you are a detective. Given clues about the whole group, can you work out what the cats caught? And sometimes the big question is different: can you *prove* that no set of mouse counts could ever fit?

Cat Café rules

- Each cat catches a whole number of mice: 1, 2, 3, ... (no cat catches 0 unless a case says so). The order of the cats does not matter.
- **Mean** = the fair share. $\text{Mean} \times \text{number of cats} = \text{total mice}$.
- **Median** = the middle cat when you line the cats up from fewest mice to most. With an even number of cats, it is halfway between the two middle cats.
- **Mode** = the count that shows up most often. In the Cat Café the mode must be the *only* most-common count, and at least two cats share it.
- **Range** = most mice caught minus fewest mice caught.
- A case is **possible** if at least one set of mouse counts fits every clue. It is **impossible** if none does.



Part 1 · Warm-up cases

Find mouse counts that fit every clue.

Case 1. Three cats. Mean 3, mode 2.

Mouse counts: Total mice = _____

Case 2. Four cats. Mean 7.5, mode 6, median 7.

Mouse counts: Total mice = _____

Case 3. Five cats. Mean 7, mode 7, range 10. There is more than one answer. How many different sets can you find?

Set A:

Set B:

Set C:

Which clue did you use first, and why?



Part 2 - Predict, then test

Before you start, circle **P** (possible) or **I** (impossible) for every case. Then test each one.

If a case is possible, write a set of mouse counts that works. If it is impossible, explain *how you know*. "I couldn't find one" is not enough. Give a reason that would convince a skeptical classmate.

#	The case	Guess	A set that works, or why none can
1	Three cats. Mean 4, mode 2.	P I	
2	Three cats. Mean 4, range 12.	P I	
3	Three cats. Mean 7, mode 10.	P I	
4	Four cats. Mean 5, mode 9.	P I	
5	Three cats. Mean 5, median 7, range 2.	P I	
6	Five cats. Mean 4, median 6.	P I	
7	Five cats. Mean 4, median 6, range 6.	P I	
8	Three cats. Mean 4.5, mode 4.	P I	

Which case surprised you most, and why?



Part 3 · Move one dial

Keep the mean at 3. Change only the mode and watch what happens.

Three cats with mean 3 have **9** mice altogether. Four cats with mean 3 have **12**. For each mode, write a set that works, or write X if it is impossible.

Mode	Three cats, mean 3	Four cats, mean 3
1		
2		
3		
4		
5		
6		

a. What is the biggest mode possible for three cats with mean 3? Use the total of 9 to explain why nothing bigger works.

b. Predict first, then test. The biggest possible mode for three cats with **mean 4** is ____, with **mean 5** is ____, with **mean 6** is ____.

c. Write a rule: for three cats with mean M , the biggest possible mode is _____. (Hint: the mode needs two cats, and the third cat needs at least 1.)

d. For four cats with mean 3, mode 5 is impossible, yet 1, 1, 5, 5 adds up to 12. Which rule stops it from working? What would change if ties were allowed?

The impossibility toolbox

- The total is not a whole number (mean $\times$ cats must be whole).
- The range is bigger than the total allows.
- The mode needs at least two cats, and two copies of it are already too many mice.
- The median forces too many big counts (half the cats are at least as big as it).
- Every clue is fine alone, but together they clash. Which ones?

Check your answers and explore more cases at milligan-mathematics.github.io/courses/253/about-average.html. The Possibility Map shows every case for a given number of cats at once.



Part 4 - Design your own

Make one case that is possible and one that is impossible but *looks* possible.

Step 1. A possible case. Secretly choose mouse counts for up to six cats, then write clues (mean, median, mode, range) that come from your set.

Secret counts:

My clues: _____ Number of cats: _____

How many different sets fit my clues? _____ (Can you find another set besides your secret one?)

Step 2. A sneaky impossible case. Change one of your clues so that no set can work, but so that each clue on its own still looks fine. Write the new case:

Why is it impossible? Write a proof that a classmate cannot argue with:

Step 3. Swap with a partner. Do not tell them which case is which! Solve each other's cases and record your verdicts.

My partner's case	P or I?	My set, or my reason

Did your partner's verdict match what you intended? If not, who was right?



Bonus - Bigger litters

The NRICH extension: mean 4, mode 1, median 2, range 10.

Keep the four clues the same and change only the number of cats. For which litter sizes is the case possible?

Cats	A set that works, or why none can
3	
4	
5	
6	

Could **100 cats** work? Describe a litter that does, or explain why none can.